INCH-POUND

MIL-PRF-1/99D
17 April 2007
SUPERSEDING
MIL-E-1/99C
11 August 1970

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBES, TRANSMITTING

TYPES 807 AND 1625 1/

Inactive for new design
after 7 March 1997.

The requirement for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Amplifier, beam power, F1 = 60 MHz, F2 = 125 MHz.

Outline - - - 16-2 (EIA).
Base
807 - - - A5-11 (low-loss phenolic).
1625 - - - A7-13 (low-loss phenolic).
Cap - - - C1-1.
Envelope - - - ST16.
Cathode - - - Coated unipotential.
Base connections:

Pin No.	- - -	1	2	3	4	5	6	7	Cap
Element									
807	- - -	h	g2	g1	k, g3 (Note 2)	h	- - -	- - -	a
1625	- - -	h	nc	g2	g1	nc	k, g3 (Note 2)	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pg2	Pp	Pi	Ehk	Modulation	Alt
Unit:	V	V dc	V dc	V dc	mA dc	mA dc	W	W	W	v	---	ft
Type 807												
Class B AF:	6.3±10%	600	---	300	120	---	3.5	25	60	135	---	10,000
Class B RF:	6.3±10%	600	---	300	80	---	2.5	25	37.5	135	---	10,000
Class C Telep:	6.3±10%	475	-200	300	83	5	2.5	16.5	40	135	Anode	10,000
Class C Teleg:	6.3±10%	600	-200	300	100	5	3.5	25	60	135	---	10,000
Test conditions:	6.3	600	-29	300	---	---	---	---	---	---	---	---
Type 1625												
Class B AF:	12.6±10%	600	---	300	120	---	3.5	25	60	135	---	10,000
Class B RF:	12.6±10%	600	---	300	80	---	2.5	25	37.5	135	---	10,000
Class C Telep:	12.6±10%	475	-200	300	83	5	2.5	16.5	40	135	Anode	10,000
Class C Teleg:	12.6±10%	600	-200	300	100	5	3.5	25	60	135	---	10,000
Test conditions:	12.6 V dc	600	-29	300	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Not required.

1/ See note 1.

AMSC N/A FSC 5960

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TABLE I. Testing and inspection.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 6)	Symbol	Limits		Unit
					Min	Max	
	First article inspection						
1236	Power oscillation (2)	Power oscillation (1); F = 60 MHz	- - -	Po	28	- - -	W
	Conformance inspection, part 1						
1231	Emission	Eb = Ec1 = Ec2 = 50 V dc (see note 3)	0.65	Is	300	- - -	mA dc
1236	Power oscillation (1)	Ec2 = 200 V dc; Rg = 10,000 ohms; Ic1 = 6 mA dc; Ib = 100 mA dc; F = 15 MHz.	0.65	Po	33	- - -	W
1256	Electrode current (1) (anode)		0.65	Ib	24	48	mA dc
1266	Total grid current	See note 3.	0.65	Ic	- - -	- 4.0	µA dc
1201	Short and discontinuity detection	See note 5.	0.4	- - -	- - -	- - -	- - -
	Conformance inspection, part 2						
1031	Low-frequency vibration	Eb = 250 V dc; Ec2 = 100 V dc; Ec1 = -10 V dc; Rp = 2,000 ohms	- - -	Ep	- - -	500	mV ac
1036	Bump	Hammer angle = 20°	- - -	- - -	- - -	- - -	- - -
1301	Heater current Type 807 Type 1625		 - - - - - -	 If If	 810 405	 990 495	 mA mA
1336	Heater-cathode leakage		- - -	Ihk	- - -	100	µA dc
1256	Electrode current (2) (anode)	Ec1 = -100 V dc	- - -	Ib	- - -	0.5	mA dc
1256	Electrode current (screen)		- - -	Ic2	0	4.0	mA dc
1266	Primary grid emission Type 807	Eg2 = 175 V ac (approx); Eb = Ec2 = 0; Ec1 = 0 to 6 V dc; Pg2 = 5W (see note 4)	- - -	Ic2	- - -	-750	µA dc
1306	Transconductance Type 1625	Eb = Ec2 = 250 V dc; Ec1 = -14 V dc	- - -	Sm	5,100	6,900	µmhos
1236	Internal insulation		- - -	- - -	- - -	- - -	- - -
1331	Direct-interelectrode capacitance	Shield No. 312 Without shield Without shield	- - -	Cgp Cin Cout	- - - 10.0 5.3	0.2 14.0 8.7	pF pF pF

See notes at end of table.

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TABLE I. Testing and inspection - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 6)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 2 - Continued						
1101	Secureness of base, cap, or insert		- - -	- - -	- - -	- - -	- - -
1105	Permanence of marking		- - -	- - -	- - -	- - -	- - -
	Conformance inspection, part 3						
- - -	Life-test provisions	Group B; Ehk = 135 V	- - -	- - -	- - -	- - -	- - -
- - -	Life-test end points (500 hours)	Total grid current and Power oscillation (1)	- - -	Ic1 Po	0 27	- 4.0 - - -	μA dc W

NOTES:

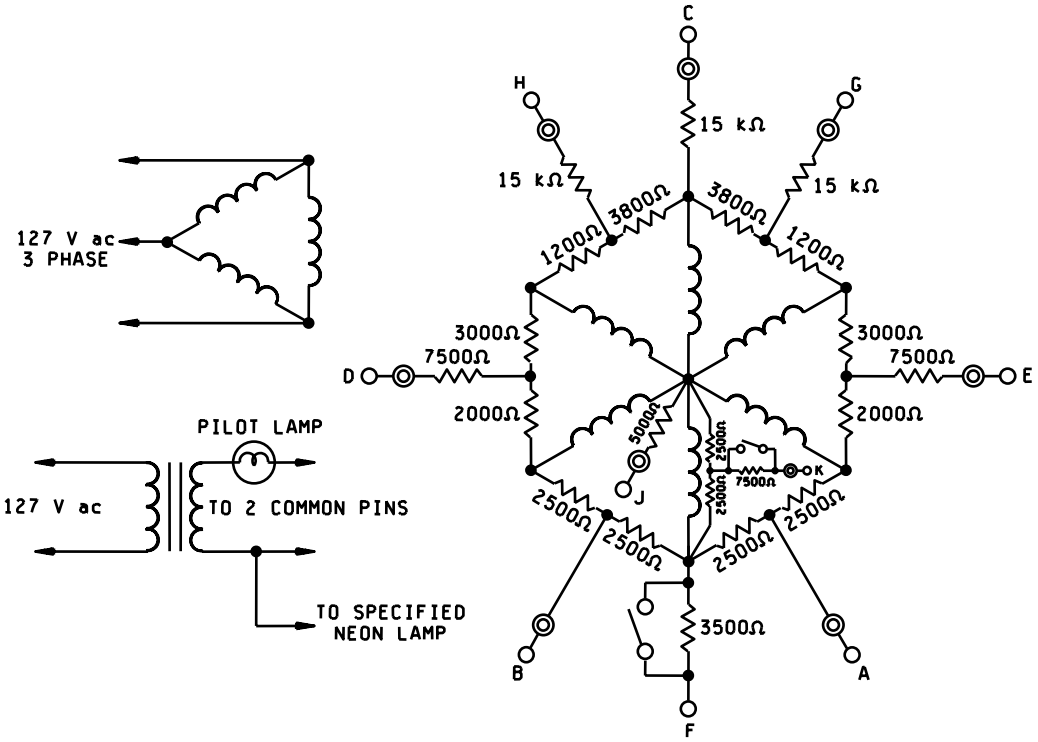
1. Tube type 5933 has been deleted from this tube specification sheet. For replacement purposes, use tube type 5933WA, MIL-E-1/852.
2. The beam forming plate lead and the cathode lead shall be individually passed through the glass stem of the tube and shall be electrically connected together only at the base pin.
3. This test to be performed at the conclusion of the holding period.
4. A protective resistor of 15,000 ohms shall be placed in series with the primary emission current meter. Grid No. 2 input power shall be calculated as 2.46 times the product of the rectified current and rectified voltage. Test duration shall be sufficient to obtain a stabilized negative Ic2 value.
5. The hexaphase short and discontinuity test may be used as an alternate equivalent test for transmitting tubes (see figure 1).
6. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

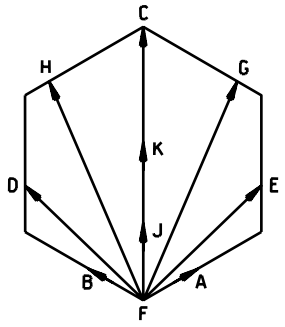
Referenced documents. In addition to MIL-PRF-1, this document references the following:

MIL-E-1/852
MIL-STD-1311

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TERMINALS	VOLTS AT SUPPLY SIDE OF NEON
FA,FB,FJ	102
FK	205
FD,FE	265
FG,FH	360
FC	410



REPRESENTATIVE TEST CIRCUIT, VECTOR DIAGRAM, AND VOLTAGES

NOTES:
1. Voltage tolerance ±15 percent. Measure voltages without a tube in the socket, using a rectifier type voltmeter (500 ohms per volt minimum) or an average-reading type VTVM, such as the Hewlett Packard 400 series. Ballantine 300 or 310, or the RCA Junior Voltohmyst, or equivalent. Peak reading meters such as the RCA Senior Voltohmyst will not agree with the above figures for voltage at the terminals. The Voltohmyst shall be isolated from ground and from the technician during the measurement, as its case is connected to its common test lead.
2. Use a hexaphase supply connected to a ring potentiometer. Each tube element (except cathode) is connected through a neon lamp to an appropriate point on the ring potentiometer.

FIGURE 1. Hexaphase short and discontinuity test.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, CG, MC, OS
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5960-2007-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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